

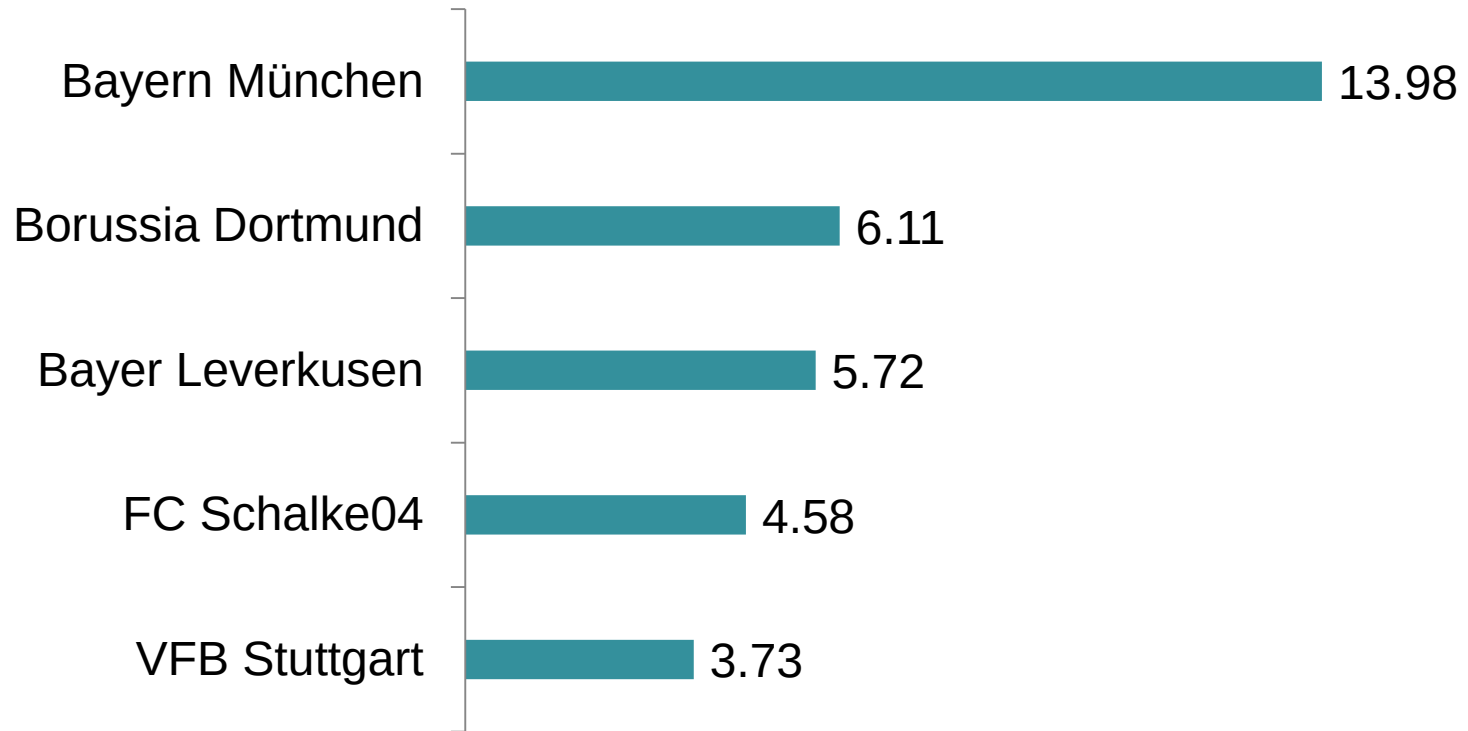


A QUESTION OF SPORTS - DOES UNDERWRITING TOP ATHLETES MEAN TOP RISKS AND UNFORESEEN CLAIMS?

Health Claims Forum, 5 December 2013

Dr. Karsten Filzmaier

Top 5 clubs of the German Bundesliga (premier league soccer)
Average market value per player in million Euro in 2012



Source: Transfermarkt

Young, healthy, fit and medically supervised ?

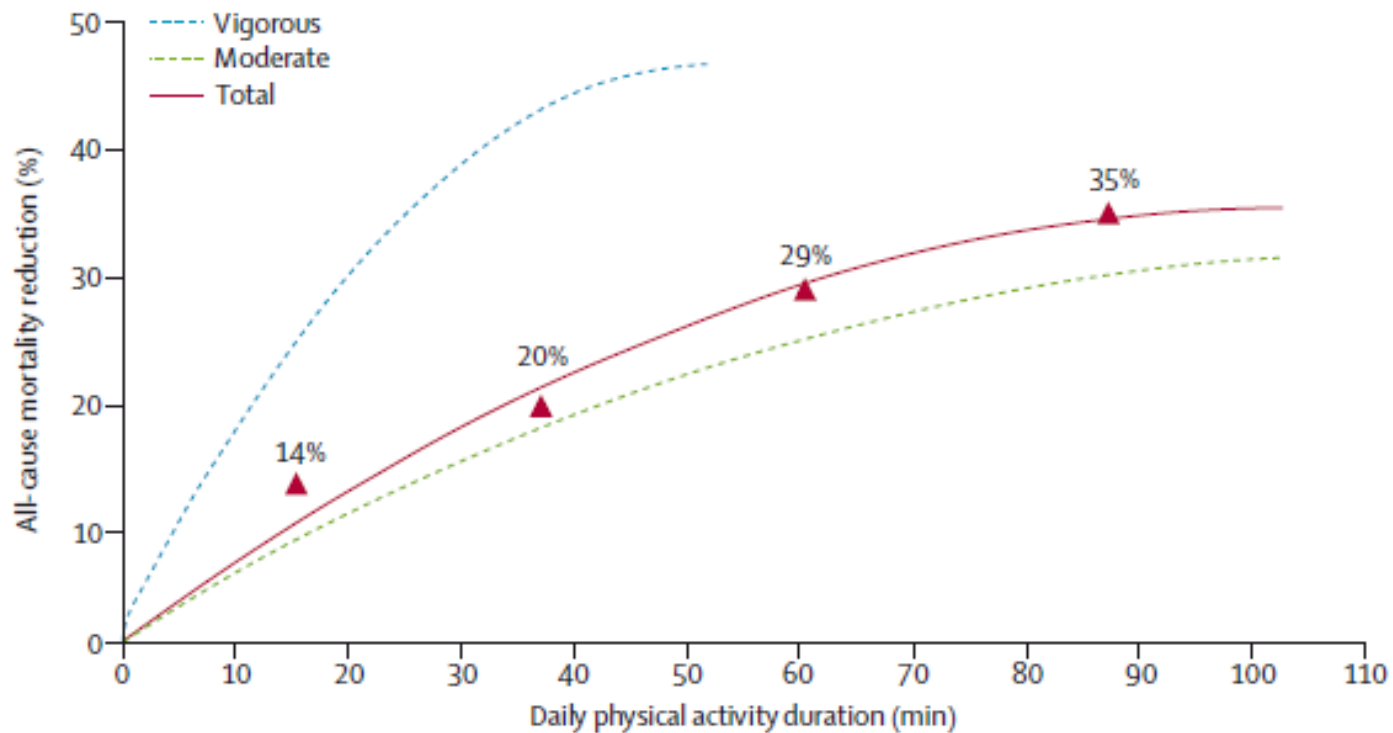
Some of the athletes suffering a cardiac arrest in the past year:

- Fabrice Muamba, a 24-year-old professional footballer from Zaire, suffered cardiac arrest during a Premier League match in England; successful resuscitation
- Piermario Morosini (26) died during an Italian football league match
- Alexander Dale Oen (26), the Norwegian world champion swimmer, died during trials for the Olympic Games, also a sudden cardiac death.
- Timo Jacobs (29), the German yachtsman, died of cardiac failure after a regatta
- Jacky Ng Kiat Kee (25), Malaysian basketball player, collapsed on the bench and was pronounced dead two hours later.
- Christian Benitez (27), Ecuadorian international footballer, never had any cardiac problems, died the day after his first game for his new club. Sudden cardiac death.

“Competitive sport begins,
where healthy sport ends”

(Bertolt Brecht)

Daily physical activity duration and reduction in all cause mortality (Taiwanese general population, n = 416 175)

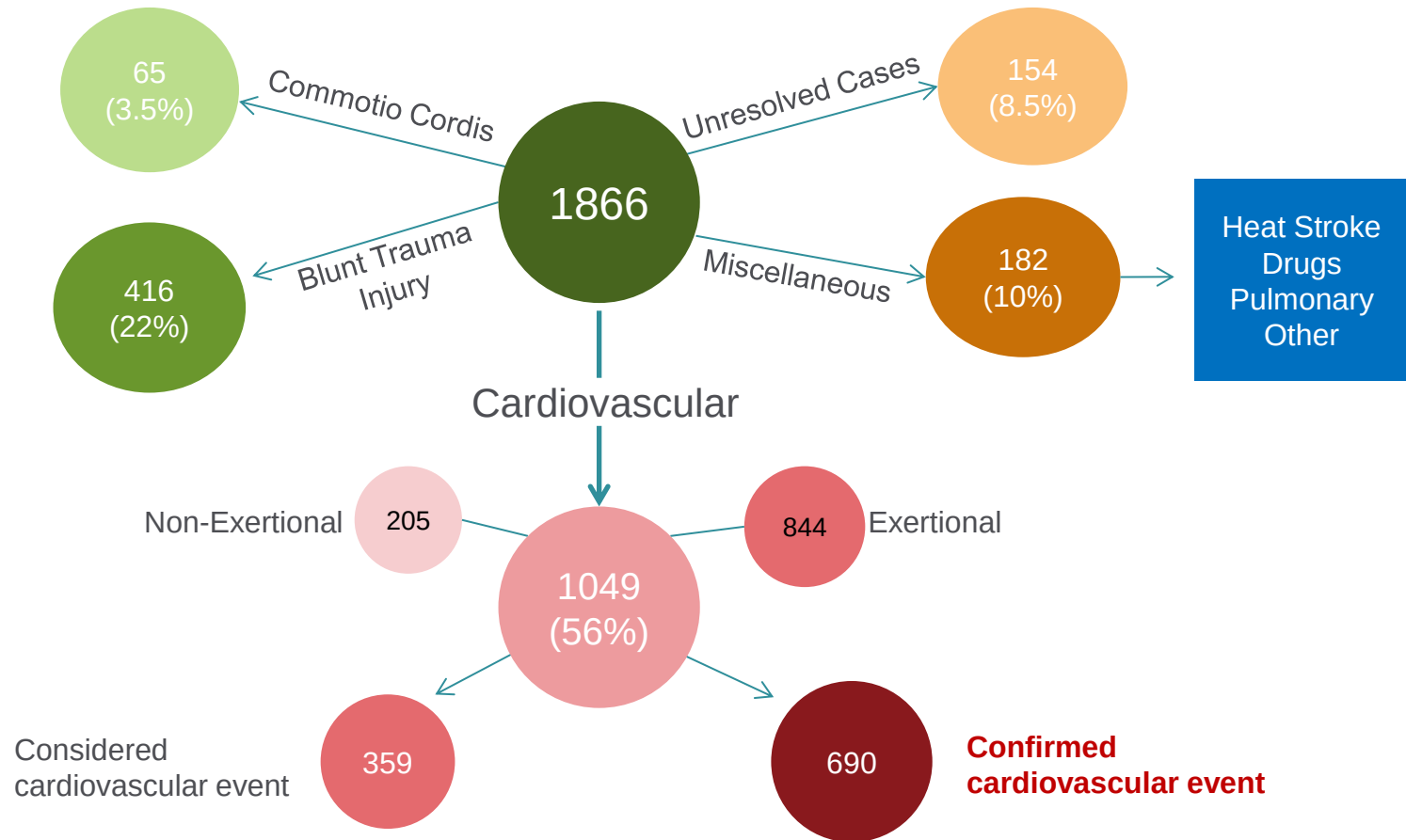


High activity level = reduced overall, cancer, cardiovascular and diabetes related mortality

Life expectancy (LE) and mortality (SMR) of athletes:

Population	♂ / ♀	N	Prognosis
Finnish athletes (e.g. soccer, ice-hockey, runners) competing internationally 1920-1965	male	1.185	LE: 4.0 years longer
Professional soccer players Italian league 1975-2003	male	5,389	SMR: 0.68
US major league baseball players 1900-1939	male	4,492	LE: 4.8 years longer
Polish Olympic athletes	male female	1,689 424	SMR: 0.5 SMR:0.73
New Zealand international rugby players since 1884	male	822	LE: same as general population
French Tour de France cyclists 1947-2012	male	786	SMR: 0.59

Causes of sudden death in 1866 competitive athletes (US registry 1980 – 2006)



Sudden Death in Athletes

Incidence Rates

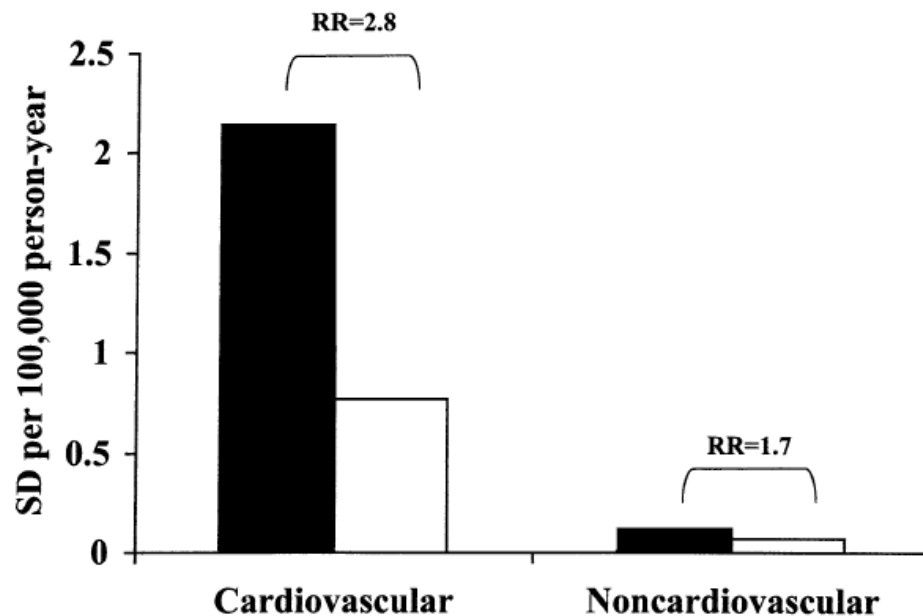


Figure 1. Incidence and relative risk (RR) of sudden death (SD) among athletes (solid columns) and non-athletes (open columns) from cardiovascular and non-cardiovascular causes. Athletes had a 2.8 RR of cardiovascular SD (confidence interval [CI] 1.9 to 3.7; $p < 0.001$), as compared with a 1.7 RR of non-cardiovascular SD (CI 0.3 to 5.7; $p = 0.39$).

Corrado D et al. *“Does Sports Activity Enhance the Risk of Sudden Death in Adolescents and Young Adults?”*

(JACC 2003; 42:1959–63)

Incidence of sudden death:

- Athletes: **2.3/100,000** per year
- Non-athletes: 0.9/100,000 per year



- Sudden death in athletes is a rare event: $\sim 0.6 - 3.0/100,000$ athletes/year
- More common during participation in basketball or soccer
- Majority of affected athletes are < 26 years of age
- Male: Female = 9:1
- Black athletes more likely to succumb to sport-related sudden death
- 80% of sudden cardiac death events during or just after physical exertion

Harmon KG et al. Incidence of Sudden Cardiac Death in National Collegiate Athletic Association Athletes *Circulation* 2011;123:1594-1600

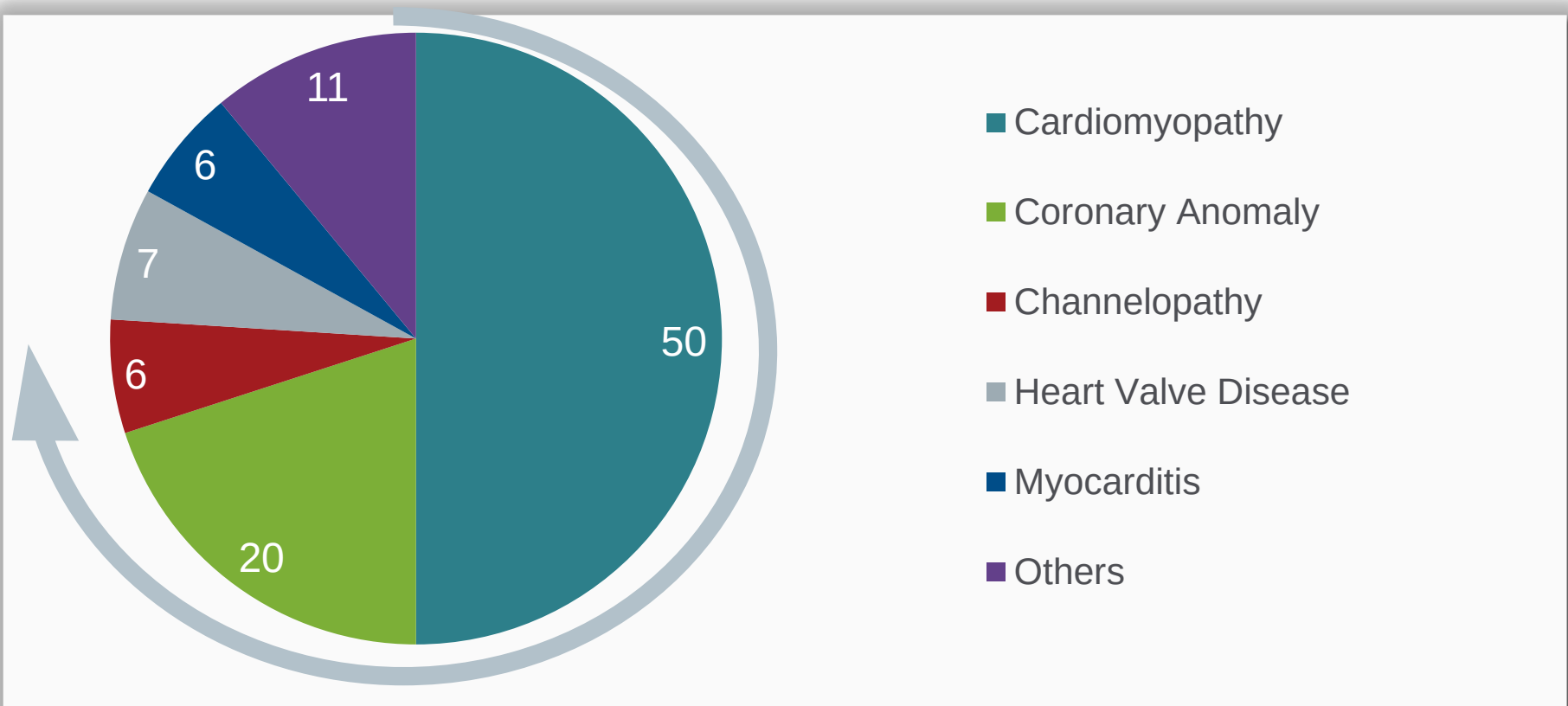
Maron BJ et al. Sudden Deaths in Young Competitive Athletes - Analysis of 1866 Deaths in the United States, 1980–2006 *Circulation* 2009;119:1085-1092

Corrado D et al. Does Sports Activity Enhance the Risk of Sudden Death in Adolescents and Young Adults? *JACC* 2003; 42:1959–63

Sudden Death in Athletes

Causes for Cardiovascular Death

Athletes suffering from sudden cardiac death:
Nearly all show an underlying (and undetected) cardiac disease post mortem

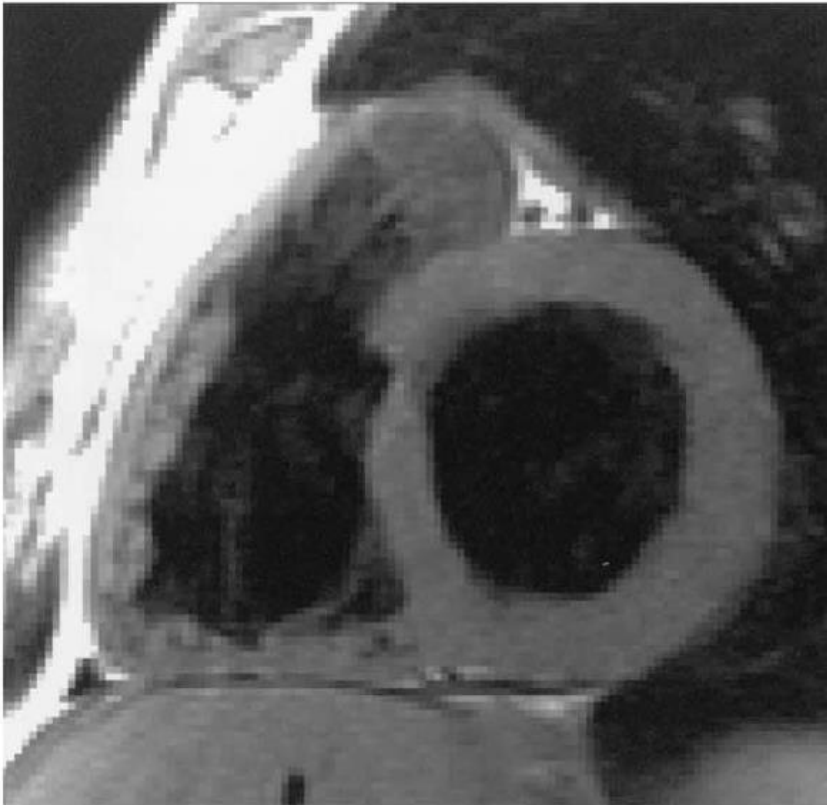


Approximately 75% of these cardiac conditions are either congenital or inherited

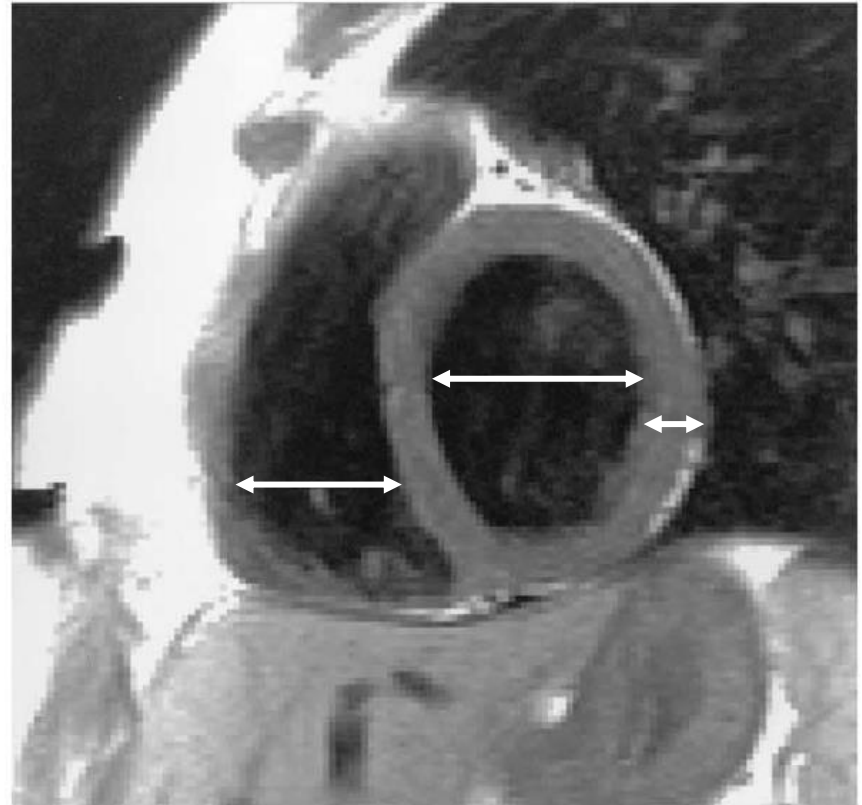


	Athlete's Heart	Cardiomyopathy
Age	15 - 40 years	First peak age 15 – 30 years
Dilation of heart chambers	✓	✓
Increase in wall thickness	✓	✓
ECG-Changes	✓	✓
Family History		✓

Endurance Athlete



Untrained Control Subject

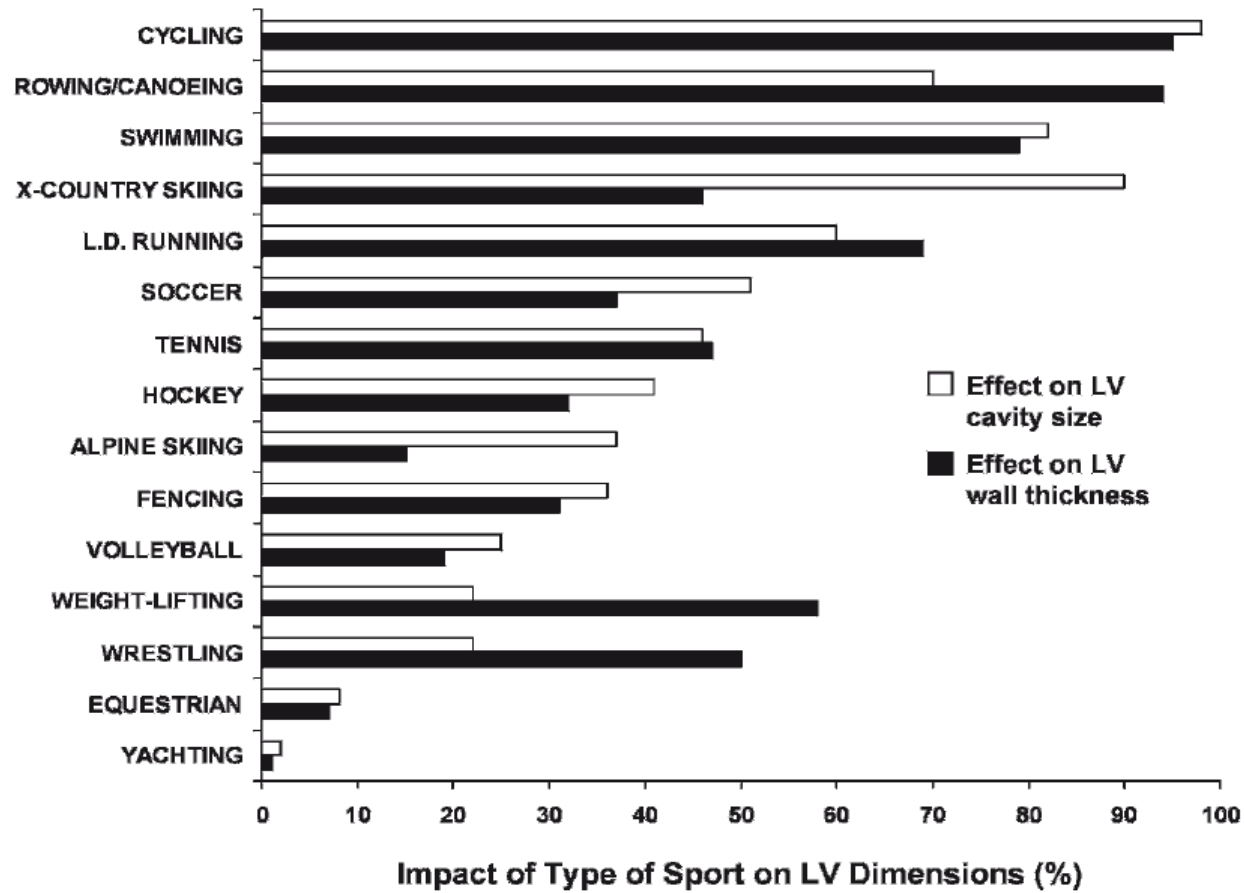


Left ventricular diastolic diameter (LVDD) = ♂ 42 – 59 mm; ♀ 39 – 53 mm

Thickness of the interventricular septum (IVSd) = ♀ and ♂ is 11 mm

Athlete's Heart

Influencing Factor: Type of Sport

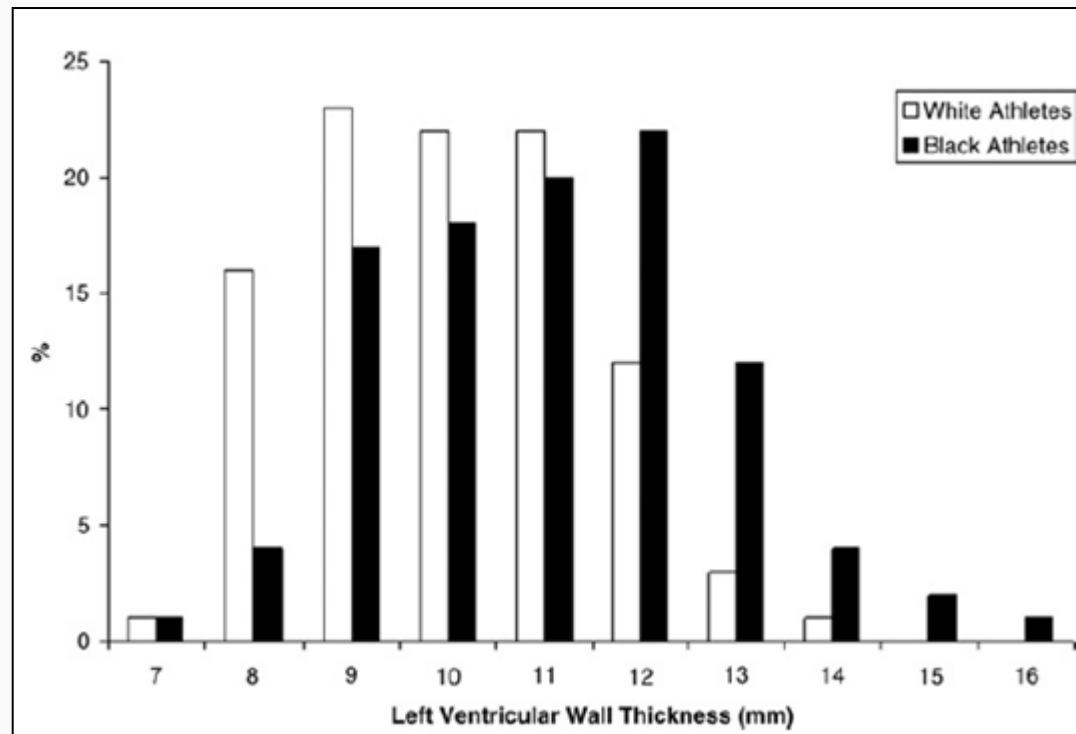


Athlete's Heart

Influencing Factor: Sex and Ethnicity

The hearts of female athletes are – compared to the general population – enlarged, but hardly ever above the upper normal limit.

Compared to white athletes, a substantial number of black athletes are developing a significant left ventricular hypertrophy (IVS ≥ 13 mm)

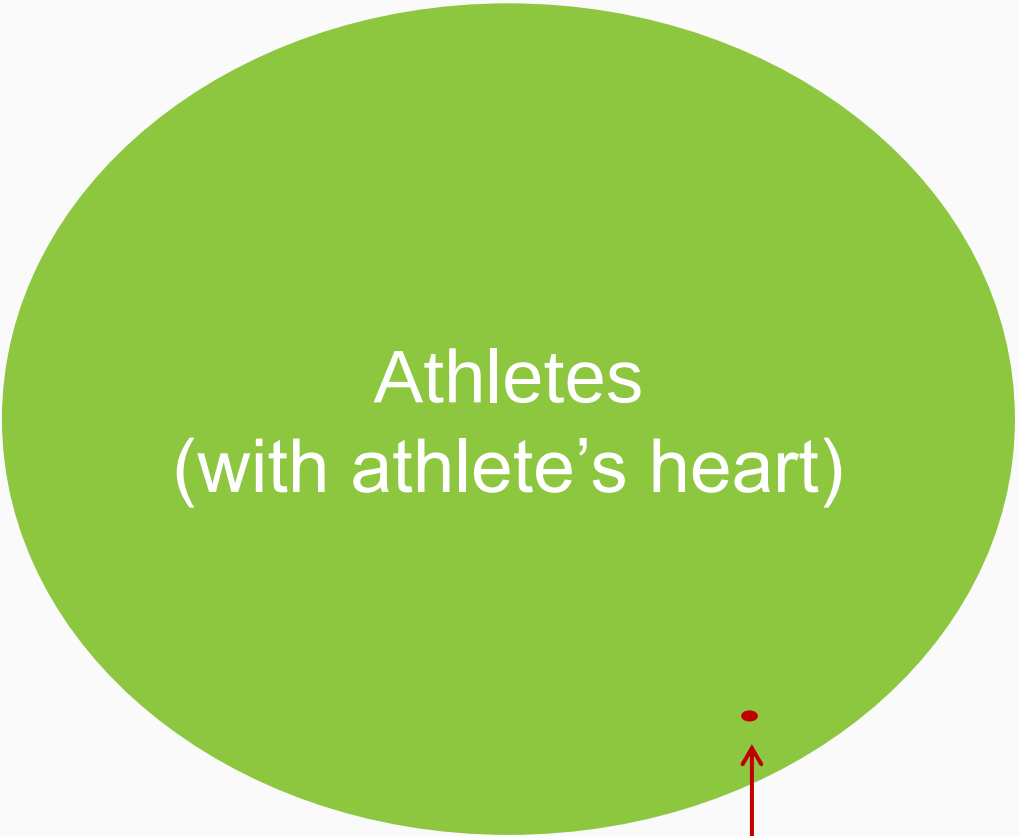


Pelliccia A et al. Athlete's Heart in Women - Echocardiographic Characterization of Highly Trained Elite Female Athletes JAMA 1996;276:211-215

Basavarajaiah S et al. Ethnic Differences in Left Ventricular Remodeling in Highly-Trained Athletes JACC 2008;51:2256-62

UNDERWRITING - STRATEGY





Athletes
(with athlete's heart)

Athletes with cardiac disease:

- ~ 0.5% of all athletes
- risk of sudden cardiac death is up to 100 times greater
- high risk that heart disease in a top athlete is misinterpreted as "athlete's heart"

Athletes with underlying
cardiac disease

Underlying cardiac diseases detectable by:

History, physical exam & ECG

Definite diagnosis possible:

- Channelopathies

Tentative diagnosis possible:

- Cardiomyopathies
- Heart valve disorders
- Marfans' disease

Echo

Definite diagnosis possible:

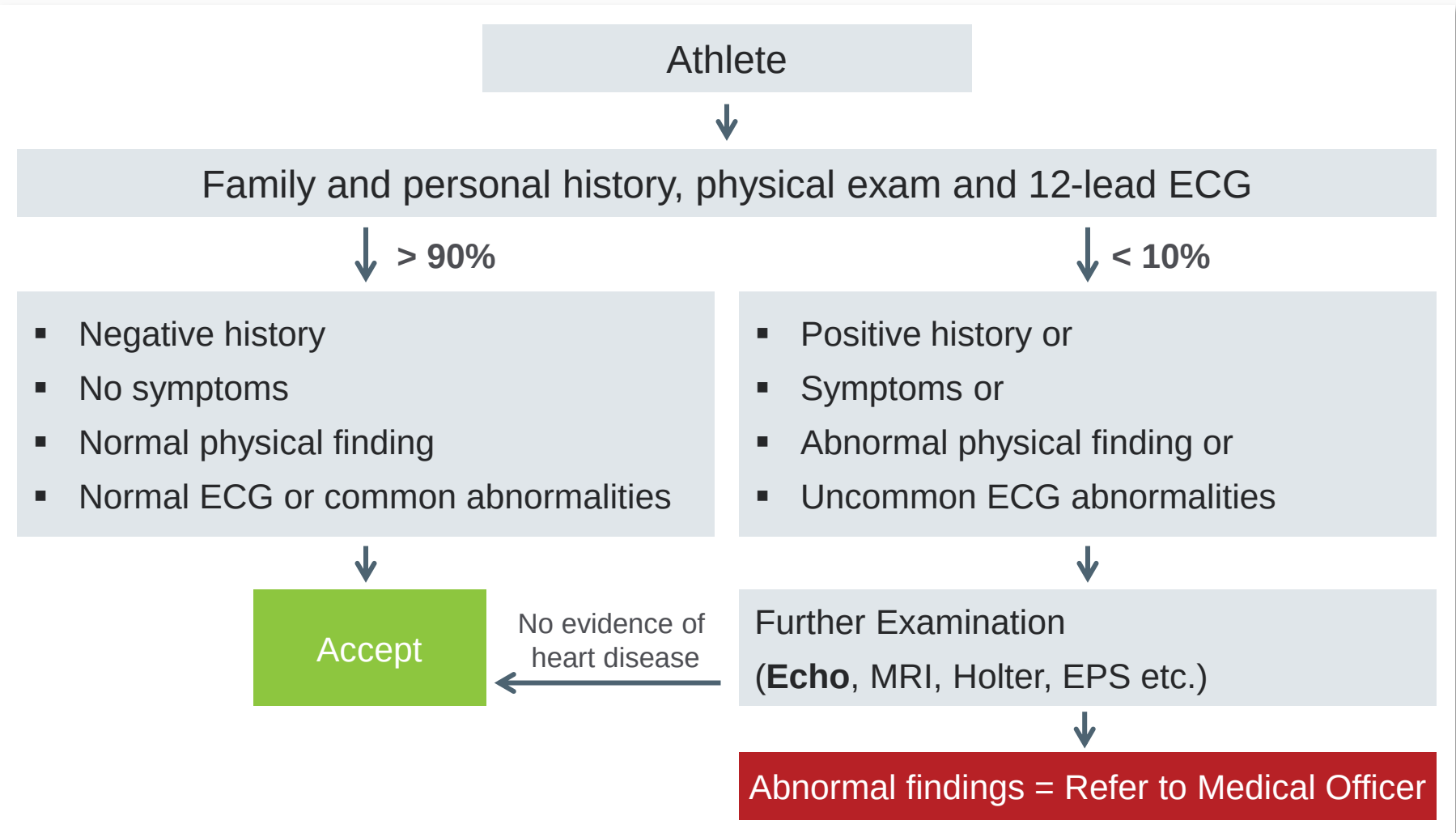
- Cardiomyopathies
- Heart valve disorders
- Marfans' disease

MRI or Cardiac CT scan

- Cardiomyopathies
- Heart valve disorders
- Marfans' disease and diseases of the Aorta
- Coronary artery disease
- Coronary anomalies
- Myocardial bridging of coronary artery

Underwriting Athletes

Algorithm for Risk Assessment

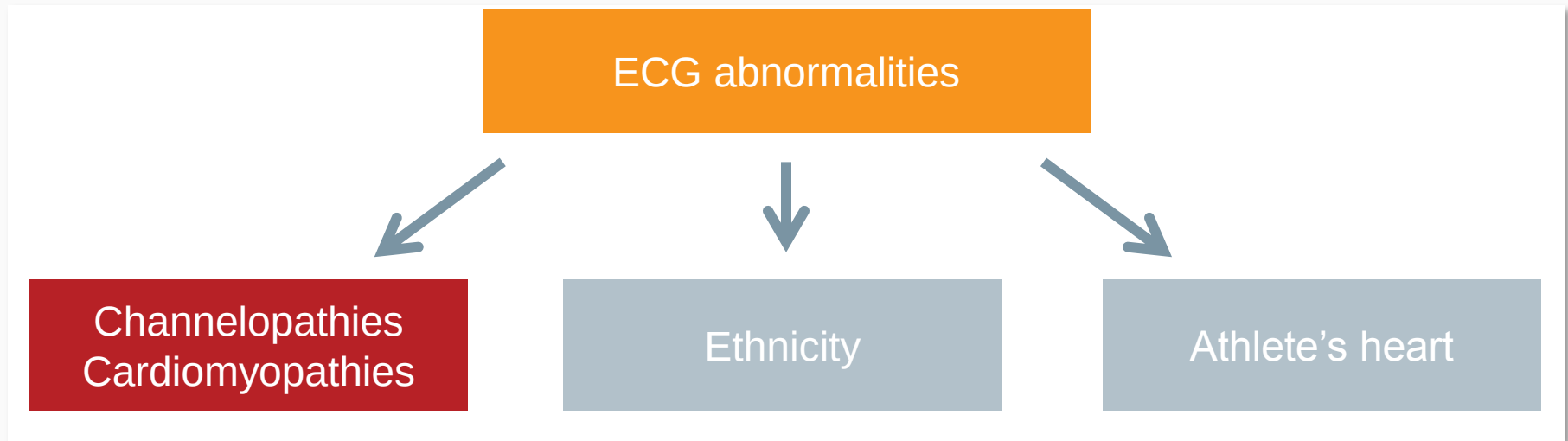


DEVELOPING EXPERTISE

- THE AFRICAN ECG -



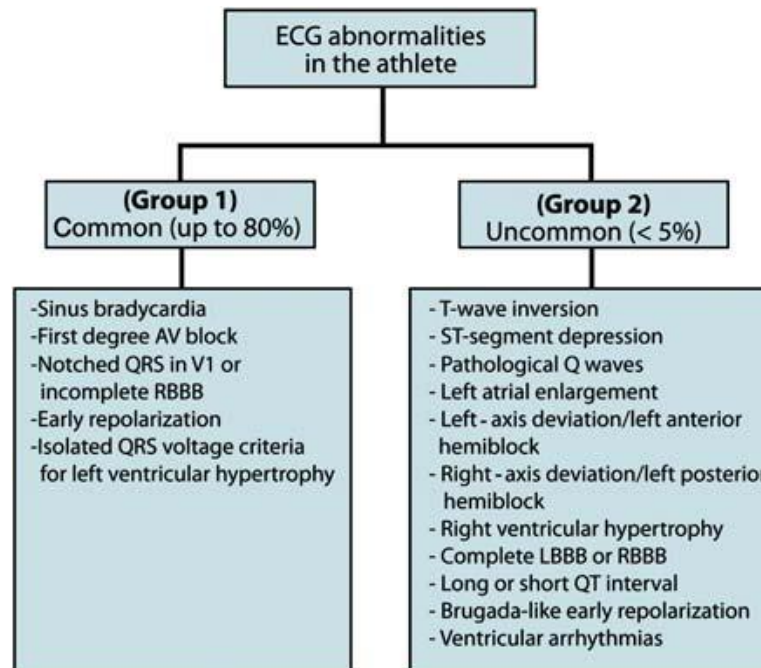
Interpreting an athlete's ECG can be challenging:



Danger of overvaluing ECG abnormalities: athlete's heart misinterpreted as cardiomyopathy
Danger of undervaluing ECG abnormalities: cardiomyopathy misinterpreted as athlete's heart

Developing Expertise

Official Criteria for the Interpretation of Athlete's ECGs



“Common” ECG abnormalities are training related and are considered as normal variants in athletes

“Uncommon” ECG abnormalities are unrelated to training and may indicate an underlying cardiac pathology

Developing Expertise

ECG and Ethnicity

ECG abnormalities in west-asian, black and caucasian male athletes:

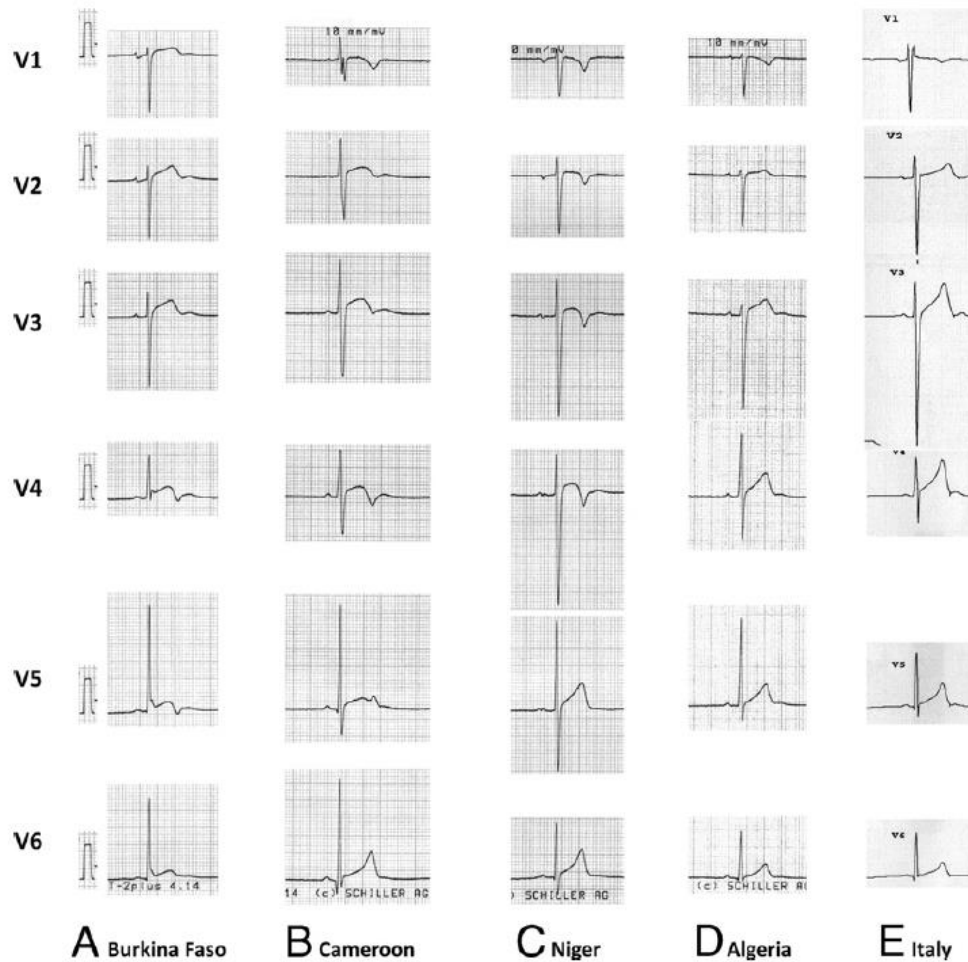
	West-Asian (n=800)	Black (n=300)	Caucasian (n=120)	West-Asian controls (n=135)
Common and training-related ECG changes	723 (90.4%)*	279 (94.6%) [†]	117 (97.6%) [‡]	105 (77.8%)
Uncommon and training-unrelated ECG changes	63 (7.9%)*	→ 53 (18%) [§]	5 (5.8%)	0 (0%)

	West-Asian (n=800)	Black (n=300)	Caucasian (n=120)	West-Asian controls (n=135)
Anterior T-wave inversion	23 (2.9%)	→ 29 (9.8%)*	0 (0%)	0 (0%)
Lateral T-wave inversion	13 (1.6%)	→ 18 (6.1%)*	0 (0%)	0 (0%)

High number of black athletes with T-wave inversion.
T-wave inversion typical ECG sign of Cardiomyopathy !

Developing Expertise

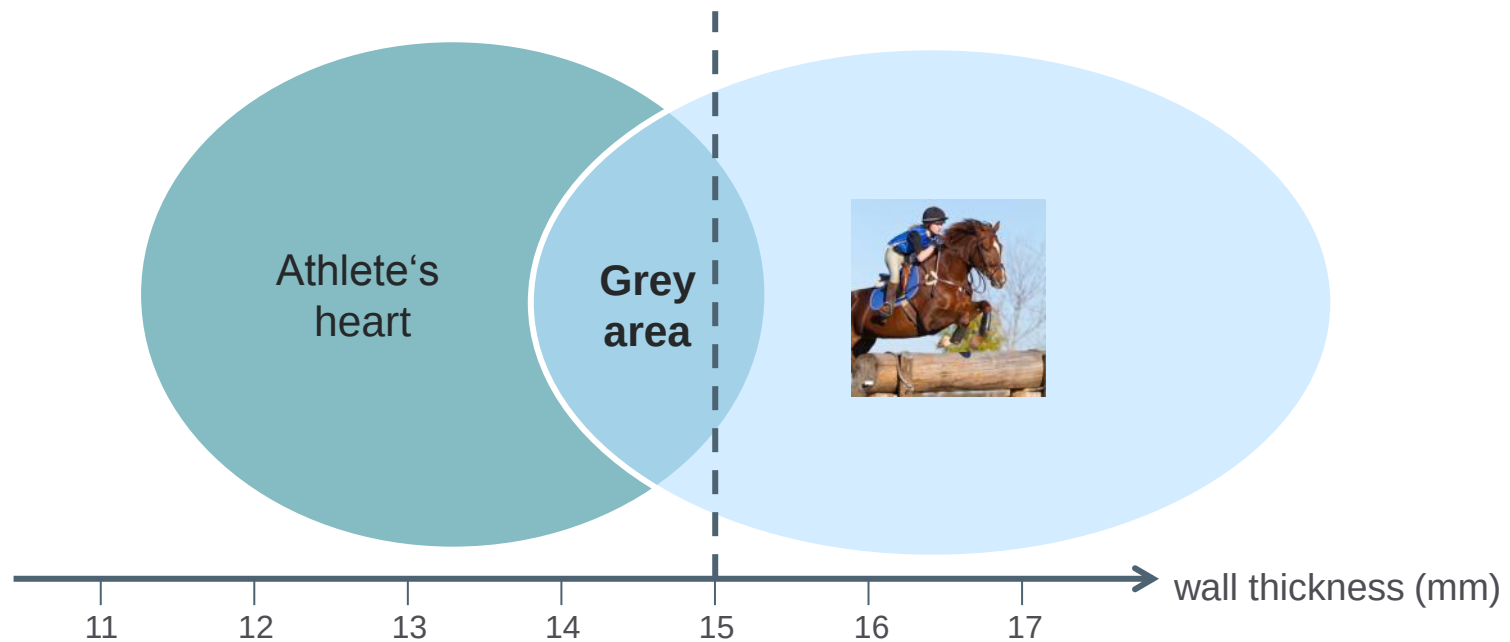
ECG and Ethnicity



Representative Differences in ECG Patterns in Adolescent African Athletes (Precordial Leads V₁ to V₆)

Case 1

Female Caucasian equestrian, 30 years
Negative history, no symptoms, normal physical finding
ECG with T-wave inversion in V2-4
Wall thickness: **15 mm**, LV dimension: 44 mm



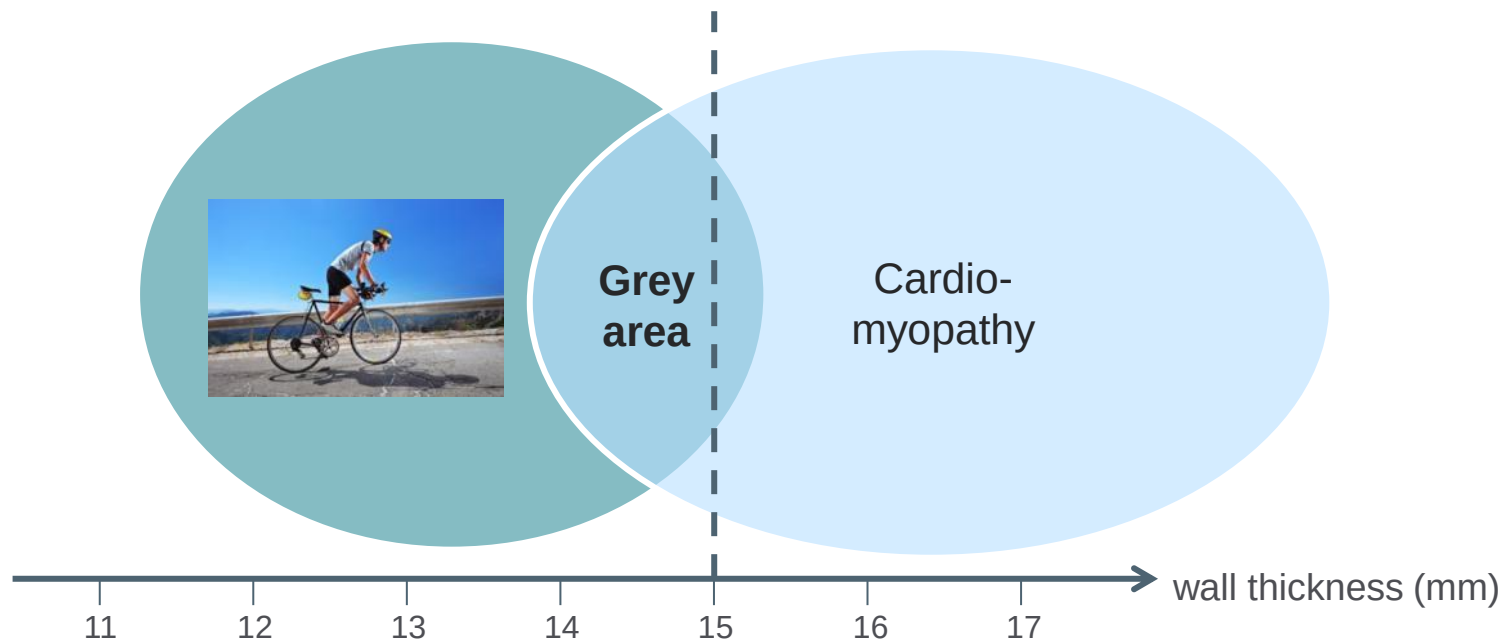
Case 2

Male Caucasian cyclist, 26 years

Negative history, no symptoms, normal physical finding

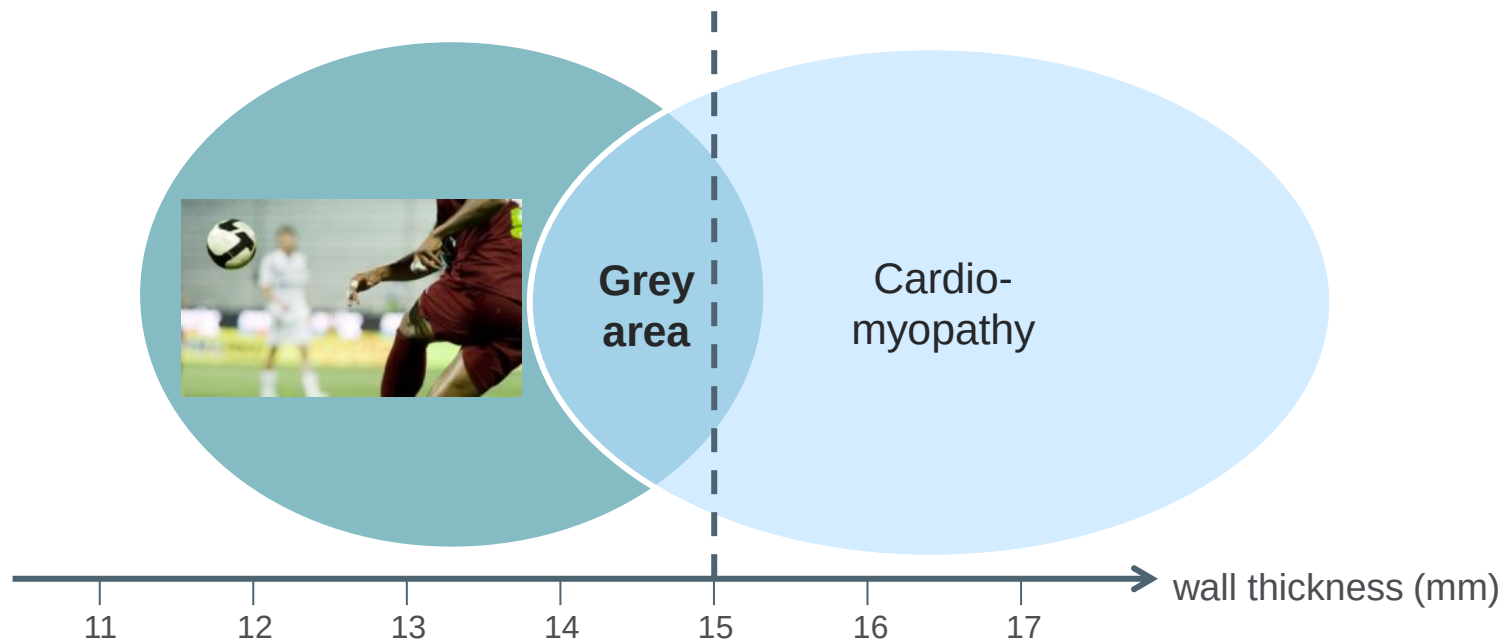
ECG: positive Sokolow index, bradycardia and AV-block °I

Wall thickness: **15 mm**, LV dimension: 63 mm



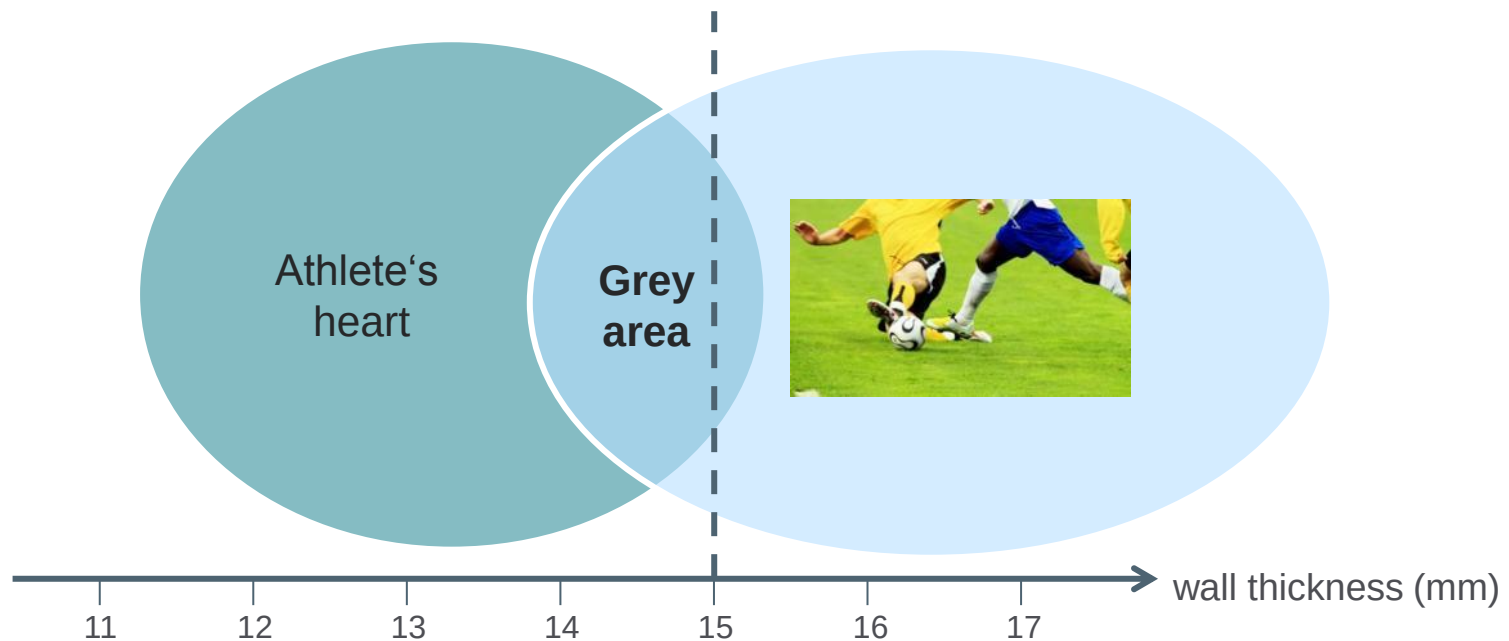
Case 3

Male black soccer player from Nigeria, 22 years
No History, no symptoms, normal physical finding
ECG: ST segment elevation + T-wave inversion V1-3
Wall thickness: **15 mm**, LV dimension: 58 mm



Case 4

Male black soccer player from Burkina Faso, 22 years
No History, no symptoms, 2/6 systolic murmur
ECG: T-wave inversion V1-3
Wall thickness: **15 mm**, LV dimension: 47 mm



What's The Claims Angle?

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Limited experience

....but impact on future life claims with?

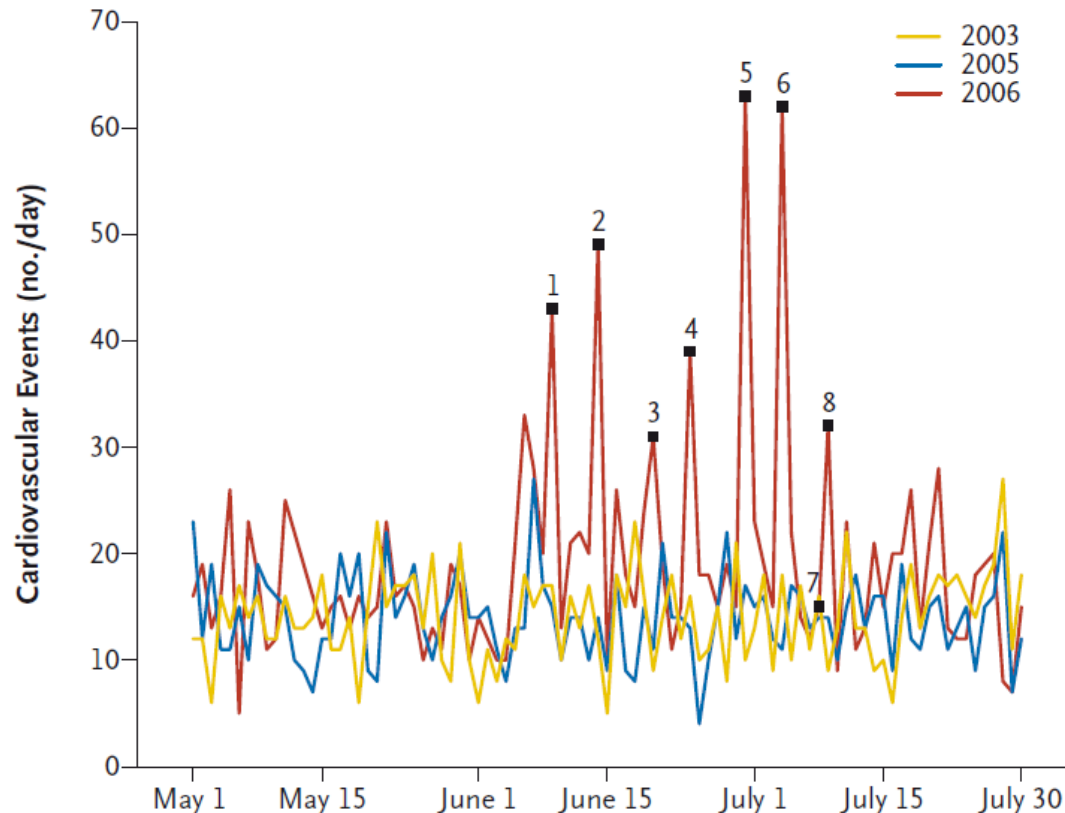
- Intensity of top levels of fitness changed in past 20 years
- Size of individual risks increasing across a greater range of sports
- Ethnic mix changing with risks seen in the UK
- Ways in which sporting risks will change in the future



Will we see more sporting claims?



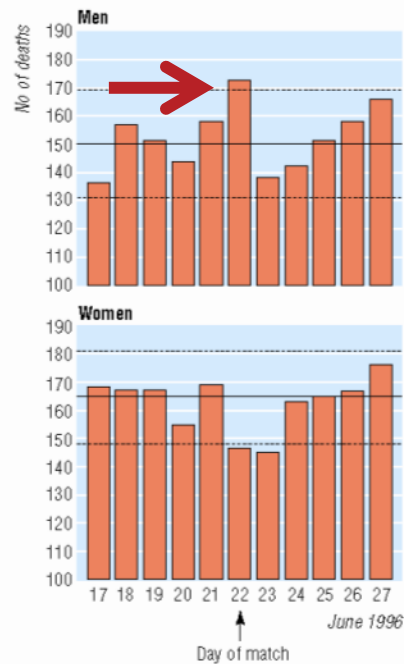
Cardiovascular Events in Bavaria between May 1 to July 31 in 2003, 2005 and 2006



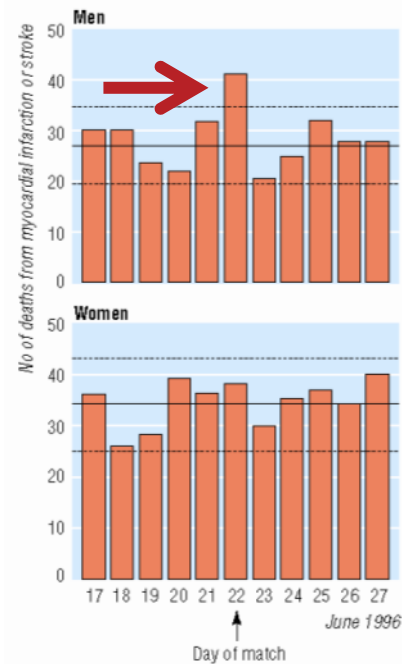
1. Germany vs. Costa Rica
2. Germany vs. Poland
3. Germany vs. Ecuador
4. Germany vs. Sweden (round of 16)
5. Germany vs. Argentina (quarter final)
6. Germany vs. Italy (semi-final)
7. Germany vs. Portugal (for third-place standing)
8. Italy vs. France (final)

Netherlands vs. France (quarter final) 4 : 5 after penalty shoot out

Death

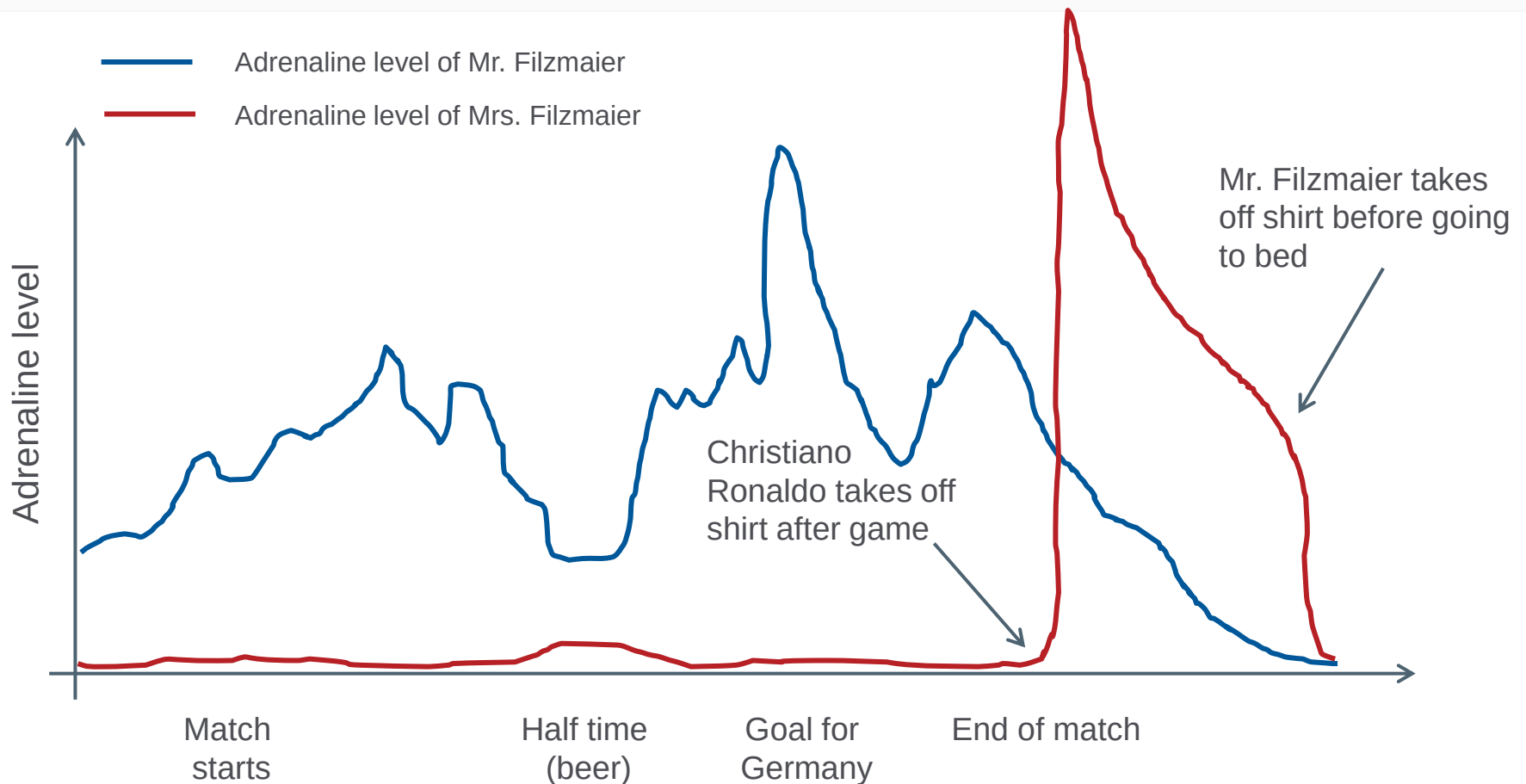


Death due to stroke or MI



Intense Emotional Reactions can Trigger Cardiac Events !

Own study: Germany vs. Portugal; Euro 2012





THANK YOU VERY MUCH FOR
YOUR KIND ATTENTION

Dr. Karsten Filzmaier



- Study: “The weekend warrior and risk of mortality”
- 8,421 healthy participants of the Harvard Alumni Health Study
Mean age 66 years, follow up between 1988 - 1993

Physical activity pattern	No of men	No of deaths	Age-adjusted relative risk
Sedentary	1,453	346	1.00
Insufficiently active	1,127	191	0.75
Weekend warrior	580	73	0.82
Regularly active	5,261	624	0.61

820 sports-related cases of sudden death in France between 2005 - 2010

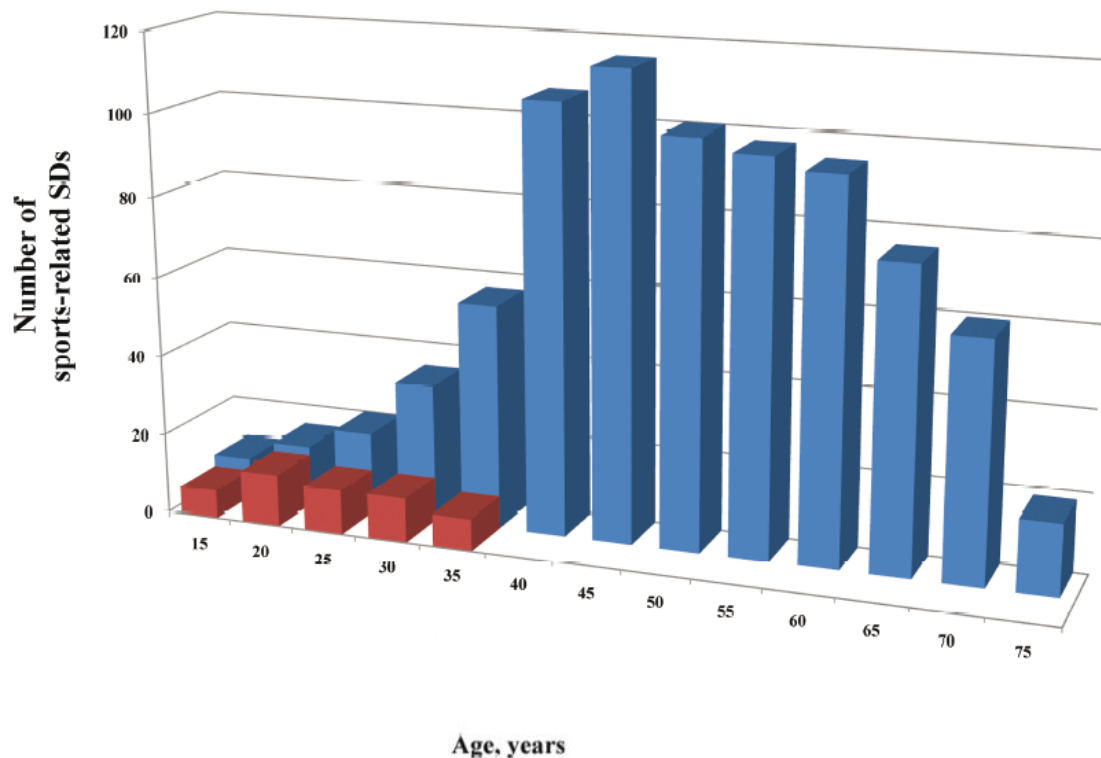
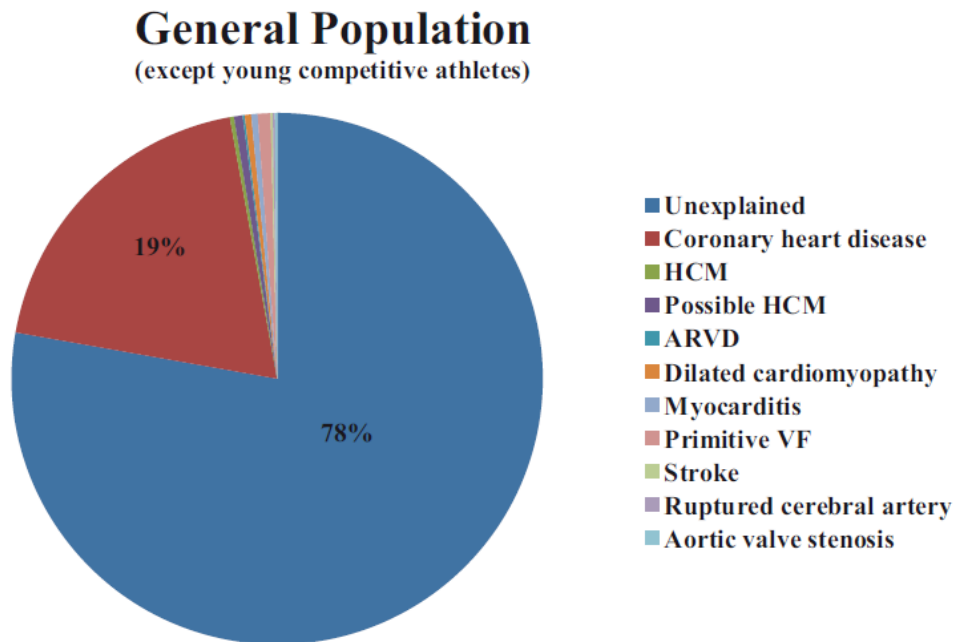


Figure 2. Distribution by age of sports-related sudden deaths (SDs) in the overall population (blue) and among young competitive athletes (red).

- 5 – 17 new cases per million population per year; 95% male
- 6% of all sports-related sudden deaths were competitive athletes
- Mean age at sudden death was 46 years

Cardiovascular abnormalities associated with sports-related sudden death in the general population



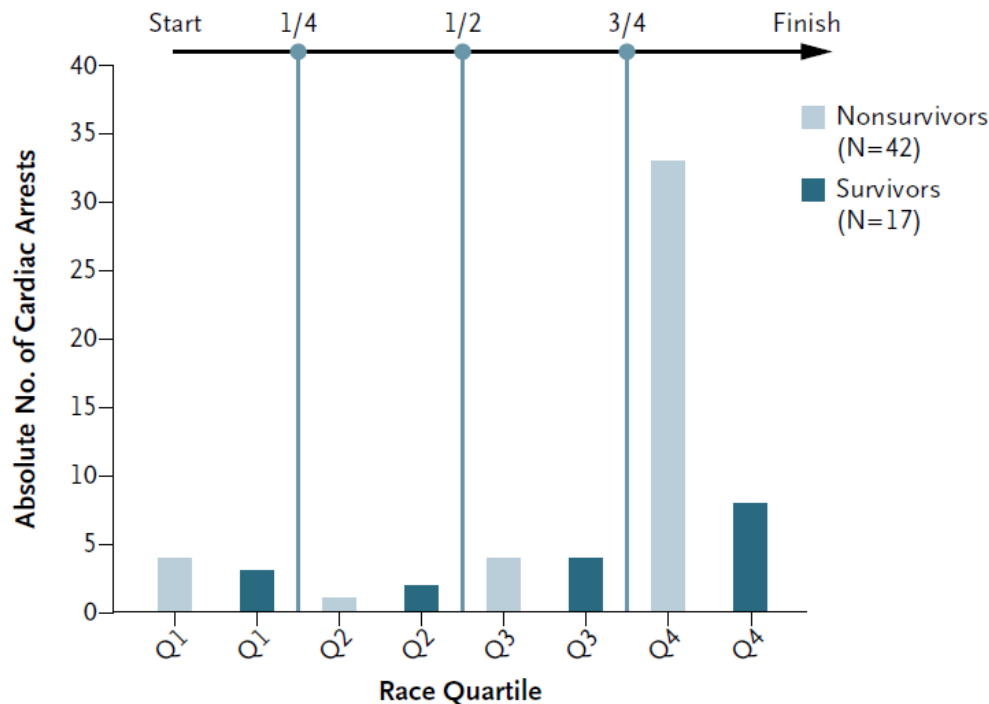
- The underlying cause of sudden death was elucidated only in 22% (203 cases)
- Among these 203 cases, coronary artery disease was the major cause of death
- Only 4 of the cases were of non-cardiac origin

Although most cases were witnessed (93%), bystander cardiopulmonary resuscitation was only commenced in 30.7% of cases

Sports-Related Sudden Death in the General Population

The Marathon Man

Location of cardiac arrest according to race quartile



- Incidence of cardiac arrest is 1/100,000 participants for marathon and 0.27/100,000 for half-marathon; 74% of all cardiac arrests fatal
- < 40 years of age = main cause HCM
- > 40 years of age = main cause coronary artery disease

In male runners, the incidence of cardiac arrest increased significantly during the last 10 years

- Prohibited substances like anabolic-androgenic steroids (AAS) or erythropoetin (EPO) are widely abused drugs (approx. 1-3 million athletes in US alone have used anabolic steroids)
- Illegal drugs = studying the health risks of anabolic steroid use is difficult and there is a paucity of well-controlled studies available

AAS side effects	AAS long term effects
Increased heart rate and blood pressure	Left ventricular hypertrophy, impaired diastolic and systolic ventricular function
Lowered HDL and increased LDL	Increased risk of myocardial infarction and sudden cardiac death
Arrhythmias	Increased risk of suicide
Nonalcoholic fatty liver disease	Increased cancer risk, esp. liver cancer and renal cell carcinoma
Depression, aggressive behaviour	
♂ Impotence, testicular atrophy, gynecomastia ♀ Masculinization, hirsutism, menstrual disorders, acne	

Table 2. Summary of Recent Studies Examining the Impact of Long-Term Term Androgenic Anabolic Steroid Use on Myocardial Function in Athletes

Reference	Year	Athletes, n	Study Design	Athlete Description	Reported Prior Steroid Exposure*	Findings Compared With Control Subjects
Nottin et al ¹¹⁵	2006	6	Cross-sectional comparison with AAS-free weightlifters	Recreational weightlifters	>2 y	No difference in systolic function; decreased diastolic function (transmitral and tissue velocities)
D'Andrea et al ¹¹⁶	2007	20	Cross-sectional comparison with AAS-free weightlifters	Top-level competitive body builders	524±91 mg/wk (31±6 wk/y over 8.9±3.8 y)	Decreased systolic function (strain, strain rate); decreased diastolic function (transmitral and tissue velocities)
Krieg et al ¹¹⁷	2007	14	Cross-sectional comparison with AAS-free weightlifters	Recreational weightlifters	817±619 mg/wk (27±11 wk/y over 8.4±4.8 y)	No difference in systolic function; decreased diastolic function (transmitral and tissue velocities)
Baggish et al ¹¹⁸	2010	12	Cross-sectional comparison with AAS-free weightlifters	Recreational weightlifters	675 [513 950] mg/wk (468 [169 520] lifetime-wk)	Decreased systolic function (LVEF, strain); decreased diastolic function (transmitral and tissue velocities)

AAS indicates androgenic anabolic steroid; LVEF, left ventricular ejection fraction.

*Numbers represent testosterone equivalences dose. Values are mean±SD when appropriate. Values in brackets are interquartile ranges.

Study: Mortality of 62 male Finnish power lifters compared to the general population

- Follow- up period 12 years
- Powerlifters placed 1st -5th in weight series 82.5-125 kg in Finnish championships during 1977-1982
- “group of subjects who are strongly suspected to have used anabolic steroids for a non-medical purpose over several years”

Results:

- 8 of the 62 power lifters (12.9%) and 34 of 1094 population controls (3.1%) have died
- The risk of death among power lifters was 4.6 times higher
- Causes of death: suicide (3), myocardial infarction (3), hepatic coma (1), lymphoma (1)

Problem: small study, “*suspected* use of AAS”, large bias